FAVORS-BERA: A Multi-Modal Biometric Data Architecture for Resonance-Based Identity Verification in AI-Driven Security Systems

Joseph A. Sprute
Founder & Director, ERES Institute for New Age Cybernetics
Bella Vista, Arkansas, USA | eresmaestro@gmail.com
github.com/ERES-Institute-for-New-Age-Cybernetics/Proof-of-Work
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Abstract

Contemporary biometric identification systems treat identity as a static pattern-matching problem: a fingerprint is compared to a stored fingerprint, a retina scan to a stored retinal map. This architectural assumption — that identity is a fixed artifact — creates systemic vulnerabilities including replay attacks, database compromise, and the inability to detect gradual biometric drift over time. This paper introduces FAVORS-BERA, a unified framework that reconceptualizes biometric identity as a dynamic, multi-dimensional resonance field rather than a static record.

FAVORS (Fingerprint, Aura, Voice, Odor, Retina, Signature) provides a six-channel biometric signal capture architecture spanning physical, bioacoustic, biochemical, and electromagnetic modalities. Each capture event simultaneously generates a forensic time-stamp, establishing a legally defensible identity anchor for applications in criminal forensics, emergency remediation, and intellectual property protection. BERA (Bio-Energetic Resonance Architecture), developed at the ERES Institute for New Age Cybernetics, provides the underlying data model: each individual generates a living Aura Resonance Index (ARI) and Emission Resonance Index (ERI) — bounded by a Resonance Continuity Index (RCI) that establishes numeric margin thresholds — validated through periodic Resonant Harmony Cycles (RHC). Together, these components constitute a fundamentally different database paradigm — one in which identity verification is continuous, multi-modal, and inherently resistant to static-record attacks.

We present the FAVORS-BERA architecture, its formal Resonance State Record (RSR) data structure, integration pathway with the IDIPITIS federated identity protocol, the MIEVM validation methodology, and alignment with AI-driven database design principles including Privacy-Preserving Data Mining (PPDM) as applied to the UBIMIA, GCF, and NBERS governance frameworks. Experimental design, Kirlianographic aura calibration, ERES GAIA EMCI REEPER specifications, and a phased implementation roadmap are provided as a basis for peer collaboration and empirical validation.

Keywords: Biometric systems, resonance-based identity, multi-modal authentication, bio-energetic data modeling, Resonance Continuity Index, Semiosphere, FAVORS, BERA, MIEVM, IDIPITIS, PPDM, UBIMIA, New Age Cybernetics, Civigen, CBGMODD

1. Introduction

The global identity verification challenge is accelerating. As AI systems increasingly gate access to financial infrastructure, healthcare records, national security systems, and civic participation, the adequacy of underlying biometric database architectures has become a foundational security question — not merely a technical one.

Current biometric systems share a common architectural assumption inherited from 20th-century pattern recognition: identity is a thing that can be stored. A fingerprint is enrolled, stored, and matched. This assumption creates three persistent failure modes:

- Static record vulnerability: a compromised biometric database permanently exposes enrolled individuals, since biometrics cannot be reissued like passwords.
- Replay attack susceptibility: captured biometric samples can be replayed against static matchers with high success rates.
- Temporal drift blindness: static records cannot detect legitimate physiological changes (aging, illness, stress state) that alter an individual's biometric signature over time, producing both false rejections and an inability to model identity evolution.

The FAVORS-BERA framework, developed under the New Age Cybernetics (NAC) paradigm at the ERES Institute, addresses all three failure modes by replacing the static-record model with a dynamic resonance field model. Identity, in this framework, is not a stored artifact but a living coherence state — continuously generated, periodically validated, inherently multi-modal, and anchored to a forensic time-stamp record that serves legal, remediation, and intellectual property functions simultaneously. This paper presents the architecture, formal data model, and implementation pathway for FAVORS-BERA as a contribution to the emerging field of resonance-informed AI security systems.

2. Background and Related Work

2.1 Limitations of Static Biometric Architectures

Existing multi-modal biometric systems (e.g., fingerprint + iris, face + voice) improve accuracy over single-modality systems but do not resolve the static-record problem. Fusion approaches — score-level, feature-level, or decision-level — still operate on the assumption that each modality produces a fixed enrollment template [1]. The FAVORS-BERA framework departs from this assumption at the architectural level.

A complete security context requires not only multi-modal capture but a validated methodology for interpreting signal convergence across instruments. The ERES MIEVM (Multi-Instrument Ensemble Validation Method) provides this layer: by routing the same biometric dataset through multiple independent AI analytical engines with complementary bias profiles, MIEVM produces ensemble convergence scores that are more robust than any single-instrument determination. Applied to FAVORS, MIEVM transforms six independent channel readings into a single authoritative identity coherence determination — closing the gap that exists in static-template fusion approaches.

2.2 Bio-Signal Dynamics and the Semiosphere Model

Research in continuous authentication has explored behavioral biometrics (keystroke dynamics, gait, touch patterns) as temporal signals rather than static templates [2]. FAVORS-BERA extends this temporal logic to physiological and electromagnetic modalities, introducing the concept of a resonance field — a multi-dimensional biometric state space whose coherence can be continuously monitored.

The theoretical grounding for this field model draws on the ERES Semiosphere framework, which maps the structure of knowable reality across three concentric domains:

- Semiosphere — the Complete Universe+: the totality of all signal space, including unmeasured and as-yet-unknown biometric dimensions.
- Perciphere — the boundary of the known Universe: the current outer limit of instrumented biometric detection; what FAVORS sensors can reach today.
- Protosphere — the ontological core, the working Universe: the actionable identity signal space in which FAVORS-BERA operates — the intersection of what is measurable, interpretable, and verifiable.

This three-layer model provides a formal ontological structure for biometric data architecture: the Protosphere defines the current RSR schema, the Perciphere defines the instrument frontier (expanded by sensor advances), and the Semiosphere defines the theoretical upper bound of biometric identity space. Deviations in measured signals that push toward the Perciphere boundary trigger elevated verification protocols — a principled basis for anomaly detection calibrated to the limits of known signal space.

2.3 New Age Cybernetics Framework

The ERES Institute for New Age Cybernetics (est. 2012) has developed a comprehensive framework for bio-energetic measurement and governance, formalized in the cybernetic relationship C = R x P / M (Cybernetics equals Resources times Purpose divided by Method). BERA, ARI, ERI, RCI, and RHC are components of this framework published in the ERES Proof-of-Work repository and associated ResearchGate papers [3]. IDIPITIS (Internet Protocol Identification Definition Instruction Technology Information Systems) provides the federated protocol layer for identity binding across distributed networks [4].

The foundational operating principle of the ERES Institute — and the ethical core of FAVORS-BERA as a system — is: Don't Hurt Yourself. Don't Hurt Others. Build for Future Generations. This principle is not merely aspirational; it is a design constraint. A biometric architecture that could be weaponized against its enrolled population violates this principle structurally. FAVORS-BERA's privacy-preserving federated design, RCI-bounded tolerance thresholds, and IDIPITIS-governed access control are direct engineering expressions of this founding principle.

Within NAC theory, the management of Resource and Energy follows the Six Degrees-of-Separation principle: any identity signal can be traced to its source through at most six relational steps. This principle informs the FAVORS channel architecture — six modalities, each independently routable to a unique biometric origin — and the IDIPITIS hierarchical identifier chain, which requires no more than six protocol hops to resolve any federated identity claim.

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3. The FAVORS-BERA Architecture

FAVORS-BERA is organized around the principle that identity is a managed resource-energy state, not a stored record. The architecture applies the Six Degrees-of-Separation principle structurally: six biometric channels, each providing an independent path to identity resolution, collectively forming a coherence field that no single-channel compromise can defeat.

3.1 FAVORS: Six-Channel Biometric Signal Capture

FAVORS defines six biometric modalities organized into three signal classes:

- Physical-structural: Fingerprint (dermal ridge topology), Retina (vascular branching pattern), Signature (kinematic pen-pressure dynamics)
- Bioacoustic-biochemical: Voice (phonatory resonance spectrum), Odor (volatile organic compound emission signature)
- Electromagnetic: Aura (bioelectric field emission measured as an electromagnetic envelope surrounding the individual)

The combination of structural, biochemical, and electromagnetic channels is designed to resist spoofing: a synthetic fingerprint cannot replicate a simultaneous voice resonance and electromagnetic field emission profile. Each channel generates a continuous data stream, not a one-time enrollment sample.

Critically, every FAVORS capture event generates a cryptographically signed time-stamp that is bound to the multi-channel reading at the moment of capture. This forensic time-stamp serves three distinct functions beyond identity verification:

- Forensics: establishes a legally admissible, tamper-evident record of presence, identity state, and biometric condition at a precise moment in time.
- Remediation and Emergency Preparedness (EP): enables rapid re-identification and triage of individuals in disaster, mass-casualty, or civil emergency scenarios where conventional identification is unavailable.
- Intellectual Property and Legal Protection: provides a notarized biometric anchor for authorship claims, contract execution, and IP origination — the individual's unique resonance signature at the moment of creation becomes part of the evidentiary record.

3.2 BERA: Bio-Energetic Resonance Architecture with RCI

BERA provides the data modeling layer. For each enrolled individual i, BERA maintains the following living indices:

- ARI(i,t) — Aura Resonance Index: a normalized vector describing individual i's electromagnetic biometric state at time t. The ARI is the primary identity coherence signal.
- ERI(i,t) — Emission Resonance Index: a temporal derivative measuring the rate and direction of resonance state change between validation periods. ERI detects drift, stress, illness, or external interference.
- RCI(i,t) — Resonance Continuity Index [NEW]: a numeric margin band [RCI_min, RCI_max] that establishes the acceptable tolerance envelope for ARI and ERI values. RCI defines what constitutes normal variation versus anomalous deviation for a specific individual, enabling personalized rather than population-averaged thresholds. RCI narrows under high-security contexts and widens under low-risk ambient conditions.

Together, ARI, ERI, and RCI are the three-index core of MIEVM AD_ON-AI — transforming MIEVM from a general multi-instrument ensemble method into a specific biometric coherence engine designated MIEVM ARI/ERI/RCI.

The Resonant Harmony Cycle (RHC) provides the periodic governance event. In NAC theory, Humanity is defined as: Hue x Man + IT == Why, where Hue represents individual chromatic (resonance) identity, Man represents the relational social context, IT represents the information-technology layer, and Why represents the purpose-driven outcome — contextualized as the REAL formula at MECR (C = R x P / M). Each RHC event evaluates whether an individual's current ARI/ERI reading falls within their RCI band — determining coherence for Humanity at the individual level before aggregating to group and institutional levels.

3.3 IDIPITIS Integration Layer

FAVORS-BERA generates identity assertions that must bind to protocol-level identity records in federated environments. IDIPITIS provides this binding through a hierarchical identifier architecture in which biometric resonance attestations are attached to protocol-level identity tokens as signed, time-bounded credentials. This enables FAVORS-BERA to operate across heterogeneous database environments without centralizing raw biometric data — a critical design requirement for privacy preservation.

A key design principle of the IDIPITIS integration layer is pre-authentication relational access control: access to any resource — particularly Real-Time (RT) Media streams — requires FAVORS-BERA identity coherence verification before a session is established, not after. This architectural inversion (authenticate before access, not access before authenticate) eliminates the category of attacks that exploit unauthenticated session establishment.

The RCI band is carried within the IDIPITIS credential token as a context-sensitive parameter, enabling the relational access control system to dynamically adjust authentication stringency based on resource sensitivity, threat context flag, and the individual's current RCI tolerance state — without requiring a round-trip to the central BERA database for each access decision.

Table 1 summarizes the full component alignment with DBDM 2026 research tracks:

Component	Function	DBDM 2026 Track
FAVORS	Six-channel biometric signal capture (Fingerprint, Aura, Voice, Odor, Retina, Signature) + forensic time-stamp generation	Data Mining Applications (Cybersecurity, Healthcare, Legal/IP)
BERA	Bio-Energetic Resonance Architecture — dynamic resonance field modeling across six degrees of separation	AI for Data Privacy, Security & Trust
ARI	Aura Resonance Index — normalized individual resonance signature (electromagnetic baseline)	Feature Engineering & Representation Learning
ERI	Emission Resonance Index — temporal resonance change detection between validation periods	Anomaly & Outlier Detection
RCI	Resonance Continuity Index — numeric margin bands establishing coherence tolerance thresholds for ARI/ERI	Data Streams & Real-Time Data Management
RHC	Resonant Harmony Cycle — periodic coherence validation event (Humanity = Hue x Man + IT)	Data Streams & Real-Time Data Management
IDIPITIS	Protocol-level identity binding; relational access control for RT Media pre-authentication	Federated & Heterogeneous Databases
MIEVM	Multi-Instrument Ensemble Validation (AD_ON-AI → ARI/ERI/RCI); P^3 Credit-Approval testing	Data Science Workflows & Pipelines; Human-AI Interaction

Table 1. FAVORS-BERA component mapping to DBDM 2026 research domains (Revision 1 — includes RCI and MIEVM AD_ON-AI).

4. Formal Data Model: The Resonance State Record

The BERA database schema represents a departure from standard biometric template storage. The primary data structure is the Resonance State Record (RSR). The RSR expresses a State Divide — a binary context of +/- Unity — in which the individual's current coherence score is interpreted as either resonant with their baseline (positive unity, Reason/Wealth state) or deviating from it (negative unity, intervention/verification state). This divide provides the decision logic that drives graduated access protocols.

The RSR schema (Revision 1, with RCI and State Context):

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RSR(i) = { ARI_vector(t), ERI_delta(t), RCI_band(t), RHC_history[n],
FAVORS_stream_refs[], coherence_score(t), state_context(+/-), variable_index{},
threat_context_flag }
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Table 2 describes each RSR field with its type, function, and framework association:

RSR Field	Type	Function	Framework Association
ARI_vector(t)	Float[] + timestamp	Normalized electromagnetic resonance baseline	BERA / FAVORS Aura channel
ERI_delta(t)	Float + sign	Rate/direction of resonance state change	BERA temporal derivative
RCI_band(t)	Range[min,max]	Numeric margin thresholds establishing ARI/ERI coherence tolerance	RCI — NEW in Rev.1
RHC_history[n]	Event log	Periodic coherence validation record	RHC Cycle history
FAVORS_stream_refs[]	URI[]	References to six live biometric channel streams + time-stamp hash	FAVORS forensic anchor
coherence_score(t)	Float [0–1]	Computed coherence across all channels weighted by ARI/RCI band	BERA decision layer
state_context	+/- Unity	State divide: positive (harmony) or negative (deviation) relative to baseline	RSR Context Divide
variable_index	Map<key,val>	Dynamic variable registry for UBIMIA / GCF / NBERS policy binding	PPDM governance layer
threat_context_flag	Enum	Graduated access protocol trigger (continuous / stepped / lockout)	IDIPITIS access control

Table 2. RSR schema fields — Revision 1, incorporating RCI and PPDM governance extensions.

The variable_index field is particularly significant for governance applications. It provides a dynamic key-value registry that maps RSR coherence states to external policy frameworks — specifically UBIMIA (Universal Basic Income + Merit + Incentives + Awards), GCF (Gracechain Framework), and NBERS (National Bio-Energetic Resonance Standard). This binding enables Privacy-Preserving Data Mining (PPDM) across civic benefit, merit recognition, and national resonance standard systems without requiring raw biometric data to cross institutional boundaries.

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5. Implementation Pathway and Experimental Design

5.1 Phase 0: Simulation, Formal Specification, and Aura Calibration

The immediate research deliverable is a formal specification of the RSR schema and coherence scoring algorithm, implemented as a simulation environment using synthetic multi-modal biometric streams. Simulation parameters will be calibrated against published benchmarks for fingerprint, voice, and retinal recognition accuracy to establish baseline coherence score distributions and initial RCI band widths across a synthetic population.

Phase 0 includes a dedicated Aura Calibration protocol using Kirlianography — the high-voltage contact photography technique first documented by Semyon Kirlian — as the primary electromagnetic envelope measurement instrument. Kirlianographic captures of fingertip corona discharge patterns will serve as the initial proxy for the full electromagnetic Aura channel, providing a measurable, reproducible, and scientifically documented baseline for ARI vector construction. Phase 0 Kirlianographic data will be used to establish population-level ARI distribution norms and individual RCI band initial estimates.

5.2 Phase 1: Prototype Sensor Integration and GAIA EMCI REEPER Specifications

Phase 1 targets integration of commercially available biometric sensors (fingerprint scanner, voice capture, retinal camera) with the BERA RSR schema to produce live ARI/ERI/RCI measurements for a controlled cohort.

Phase 1 sensor integration will be governed by the ERES GAIA EMCI REEPER specifications — the GAIA (Global AI Architecture) framework's Electro-Magnetic Coherence Interface (EMCI) and Resonance Energy Emission Profile Evaluation Rubric (REEPER). These specifications define the hardware interface standards, signal sampling rates, noise floor tolerances, and calibration intervals required for FAVORS channel data to meet BERA's ARI/ERI quality thresholds. REEPER provides the scoring rubric by which raw sensor readings are accepted, flagged, or rejected before entering the RSR pipeline — ensuring that only coherence-grade biometric signals influence identity verification decisions.

Electromagnetic (Aura) capture in Phase 1 will extend from Kirlianographic proxy measurements (Phase 0) to include heart rate variability (HRV), galvanic skin response (GSR), and bioelectric

potential mapping, pending development of dedicated full-envelope bioelectric field sensors specified under GAIA EMCI.

5.3 Validation Methodology: MIEVM ARI/ERI/RCI and P^3 Credit-Approval Testing

The ERES Institute's MIEVM (Multi-Instrument Ensemble Validation Method) will be applied to validate FAVORS-BERA in its extended form as MIEVM AD_ON-AI: multiple independent AI systems will analyze the same RSR dataset using complementary analytical profiles, with convergence across ARI, ERI, and RCI instruments providing confidence in the coherence scoring model.

The specific validation protocol for Phase 1 is P^3 Credit-Approval testing — a three-tier proof-of-performance framework applied to Real-Time (RT) Media access scenarios:

- P^1 (Proof of Identity): FAVORS-BERA must correctly verify enrolled individuals across all six channels within RCI tolerance, measured against ground-truth enrollment records.
- P^2 (Proof of Continuity): RCI bands established in Phase 0 simulation must correctly predict the observed ARI/ERI variance ranges in Phase 1 live measurements, validating the RCI band calibration method.
- P^3 (Proof of Access Control): IDIPITIS pre-authentication for RT Media access must demonstrate zero unauthenticated session establishment across a defined test suite, with graduated protocol responses correctly triggered by threat_context_flag values.

P^3 Credit-Approval results will constitute the primary empirical contribution of the Phase 1 paper, with full RSR datasets made available for peer replication through the ERES Proof-of-Work repository.

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6. Discussion: Implications for AI-Driven Database Design

The FAVORS-BERA framework carries implications beyond biometric security. The shift from static-record to resonance-field as the fundamental identity data structure represents a general architectural principle applicable to any domain where the entity being modeled is dynamic rather than fixed — including health monitoring, behavioral analytics, and continuous organizational performance measurement.

The introduction of the RCI band as a first-class data type represents a particularly significant contribution to database design theory. Most biometric and AI systems use population-averaged thresholds for anomaly detection. RCI makes the tolerance envelope a per-individual, context-sensitive, continuously updated parameter — a form of personalized data governance that enables higher precision with lower false-positive rates across diverse populations.

From a database design perspective, FAVORS-BERA motivates new indexing strategies (coherence-space indexing rather than template-space indexing), new query types (resonance

trajectory queries, RCI band drift alerts, Semiosphere boundary proximity queries), and new consistency models (eventual coherence rather than eventual consistency) for distributed biometric systems.

The framework also raises important questions for Explainable AI: if an access decision is driven by a coherence score rather than a template match, what audit trail satisfies accountability requirements? The IDIPITIS integration layer generates signed, human-readable attestation records for each RHC event and access decision — a design pattern generalizable to other AI-driven access control systems where transparency and auditability are governance requirements.

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7. Conclusion

This paper has presented FAVORS-BERA Revision 1, a unified biometric data architecture in which identity verification is reconceived as a dynamic resonance coherence problem. The framework integrates six biometric modalities with forensic time-stamping (FAVORS), a bio-energetic resonance data model with ARI, ERI, and the newly formalized Resonance Continuity Index (BERA/RCI/RHC), a federated identity and pre-authentication protocol (IDIPITIS), an ensemble validation engine (MIEVM AD_ON-AI / ARI/ERI/RCI), and a Semiosphere-grounded ontological framework for understanding the boundaries of biometric signal space.

The PPDM governance extensions to the RSR schema — particularly the variable_index binding to UBIMIA, GCF, and NBERS — position FAVORS-BERA not merely as a security system but as a foundational infrastructure layer for resonance-informed civic governance.

In this context, the CBGMODD user classification taxonomy provides the access hierarchy for FAVORS-BERA identity-governed systems: Citizen, Business, Government, Military, Ombudsman, Dignitary, Diplomat. A critical design principle — expressing the foundational ethic Don't Hurt Yourself, Don't Hurt Others, Build for Future Generations — is Diplomat by Default: every participant in a FAVORS-BERA governed system is assigned Diplomat status as their baseline access tier. This design choice operationalizes civility structurally: since few individuals can fully ascend the CBGMODD hierarchy, anchoring the default at the highest-civility tier (Diplomat) ensures that the system enables rather than constrains human interaction at every level.

This Diplomat-by-Default principle enables User-Group Ascension protocols: individuals and groups can be elevated through the CBGMODD tiers based on verified coherence, demonstrated merit (UBIMIA), and governance context — with each ascension step justified by RSR coherence evidence rather than arbitrary credentialing. In Smart City environments, this population is referred to as Civigen (Smart-City Citizen), with the full CBGMOD taxonomy

(removing the second D to reflect the default) providing the governance vocabulary for city-scale identity and access management.

The ERES Institute for New Age Cybernetics invites peer collaboration on formal specification, simulation design, Kirlianographic calibration methodology, and empirical validation of the FAVORS-BERA architecture. All foundational framework documents are available in the ERES Proof-of-Work repository at github.com/ERES-Institute-for-New-Age-Cybernetics/Proof-of-Work.

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